

10 (a) State Wien's displacement law.

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..... [2]

(b) The Sun has a radius of $6.96 \times 10^8 \text{ m}$ and a surface temperature of 5770 K .

The Earth is at a distance of $1.52 \times 10^{11} \text{ m}$ from the Sun.

(i) Calculate the luminosity of the Sun.

luminosity = W [2]

(ii) Determine the radiant flux intensity at the Earth of radiation emitted by the Sun. Give a unit with your answer.

radiant flux intensity = unit [3]

- (iii) Star X has a wavelength of maximum emission of $0.581\lambda_s$, where λ_s is the wavelength of maximum emission from the Sun.

Calculate the surface temperature of star X.

surface temperature = K [1]

[Total: 8]